

August-Roche-Magnus Formula for Dew Point and Saturation Vapor Pressure Estimation

Abstract & Introduction

The August-Roche-Magnus (ARM) formula is an empirical equation widely used in meteorology, environmental monitoring, and Internet of Things (IoT) applications to estimate the saturation vapor pressure and dew point temperature of air. It provides a computationally efficient and accurate approximation over the temperature range commonly encountered in atmospheric conditions. Due to its simplicity and reliability, the formula is extensively implemented in embedded systems such as ESP32, Arduino, Raspberry Pi, and commercial weather stations. In the proposed SIH project, the ARM formula is employed to calculate dew point and moisture-related atmospheric parameters from real-time temperature and relative humidity measurements, enabling intelligent environmental monitoring and early warning capabilities.

Humidity plays a significant role in determining environmental comfort, weather conditions, and the risk of condensation. The dew point temperature represents the temperature at which air becomes saturated with water vapor, causing condensation to occur. Accurate estimation of dew point is essential in applications such as smart agriculture, cold storage management, industrial monitoring, HVAC systems, and weather forecasting.

The August-Roche-Magnus formula is one of the most widely accepted empirical approximations for calculating saturation vapor pressure. Compared with more complex thermodynamic equations, it requires minimal computational resources while maintaining excellent accuracy for typical atmospheric temperatures, making it ideal for IoT-based monitoring systems.